

In [7]:

```
import numpy as np

list1=[1,2,3,4,5,6,7,8,9,10,11,12]
arr=np.array(list1)
print(arr)
arr2d=arr.reshape(2,6)
print(arr2d)
arr3d1=arr.reshape(2,6,1)
print(arr3d1)
arr3d2=arr.reshape(2,2,3)
print(arr3d2)
```

```
[ 1  2  3  4  5  6  7  8  9 10 11 12]
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
[[[ 1]
   [ 2]
   [ 3]
   [ 4]
   [ 5]
   [ 6]]

 [[ 7]
   [ 8]
   [ 9]
  [10]
  [11]
 [12]]]]
[[[ 1  2  3]
   [ 4  5  6]]

 [[ 7  8  9]
  [10 11 12]]]
```

In [9]:

```
arr3d2=arr3d2.reshape(4,3)
print(arr3d2)
```

```
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
```

In [10]:

```
arr3d2=arr3d2.reshape(12)
print(arr3d2)
```

```
[ 1  2  3  4  5  6  7  8  9 10 11 12]
```

In [11]:

```
list1=[1,2,3]
list2=[4,5,6]
list3=[7,8,9]
list4=[11,12,13]
arr=np.vstack((list1,list2,list3,list4))
print(arr)
```

```
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [11 12 13]]
```

```
In [13]: arr=np.delete(arr,2,axis=1)
print(arr)
```

```
[[ 1  2]
 [ 4  5]
 [11 12]]
```

```
In [18]: import numpy as np

# Encoded Data (20 Rows, 5 Columns)
# Columns: [houseid, housetype, price, units, location]
# housetype: Flats=0, Apartments=1, Independent=2
# location: Delhi=0, Gurgaon=1, Noida=2, Faridabad=3, Greater Noida=4, Ghazia

real_estate_data = np.array([
    [1, 0, 45, 10, 0],
    [2, 1, 60, 8, 1],
    [3, 2, 85, 4, 2],
    [4, 0, 50, 12, 3],
    [5, 1, 70, 6, 4],
    [6, 2, 90, 3, 5],
    [7, 0, 55, 9, 0],
    [8, 1, 68, 7, 2],
    [9, 0, 49, 11, 1],
    [10, 2, 88, 5, 4],
    [11, 0, 53, 10, 5],
    [12, 1, 62, 9, 3],
    [13, 0, 47, 13, 0],
    [14, 2, 83, 4, 1],
    [15, 1, 69, 8, 2],
    [16, 0, 52, 12, 3],
    [17, 1, 71, 7, 4],
    [18, 2, 95, 2, 5],
    [19, 0, 54, 11, 2],
    [20, 1, 73, 6, 0]
])

rev=np.sum(real_estate_data[:,2])
print(rev)
total=real_estate_data[:,2]*real_estate_data[:,3]
print(total)
print(sum(total)*100000)
```

```
1319
[450 480 340 600 420 270 495 476 539 440 530 558 611 332 552 624 497 190
 594 438]
943600000
```

```
In [22]: flats=real_estate_data[real_estate_data[:,1]==0]
print(flats)
totalflats=flats[:,2]*flats[:,3]
print(sum(totalflats))
```

```
[[ 1  0 45 10  0]
 [ 4  0 50 12  3]
 [ 7  0 55  9  0]
 [ 9  0 49 11  1]
 [11  0 53 10  5]
 [13  0 47 13  0]
 [16  0 52 12  3]]
```

```
[19  0 54 11  2]]
4443
```

```
In [31]: gf=real_estate_data[real_estate_data[:,3]>=10]
print(gf)
```

```
[[ 1  0 45 10  0]
 [ 4  0 50 12  3]
 [ 9  0 49 11  1]
 [11  0 53 10  5]
 [13  0 47 13  0]
 [16  0 52 12  3]
 [19  0 54 11  2]]
```

```
In [35]: gf=gf[gf[:,4]==0]
print("Properties in Delhi with More than 10 Units:\n\n",gf)
```

Properties in Delhi with More than 10 Units:

```
[[ 1  0 45 10  0]
 [13  0 47 13  0]]
```

```
In [37]: import pandas as pd
s = pd.DataFrame([10, 20, 30,40])
print(s)
```

```
   0
0  10
1  20
2  30
3  40
```

```
In [41]: dictr={"ProductID": [1001,1002], "ProductName": ["Samsung", "No Product"], "ProductPrice": [50000, 150000], "Stock": ["In Stock", "Out of Stock"]}
s1=pd.DataFrame(dictr)
print(s1)
```

	ProductID	ProductName	ProductPrice	Stock
0	1001	Samsung	50000	In Stock
1	1002	No Product	150000	Out of Stock

```
In [ ]: print(s1[[""]])
```